■ CockpitSentinel: Master Model Evaluation Report

Department of Computer Science & Engineering — Walchand College of Engineering, Sangli

Executive Summary: CockpitSentinel integrates MediaPipe face mesh geometry and a custom fine-tuned YOLO11s unified distraction detector. This report presents complete empirical evaluations across 5 distinct benchmarks: In-Cabin Distraction (v1 & v2), YawDD Fatigue Videos, Kaggle NTHU, and Kaggle State Farm DDD.

1. Primary Production Model Metrics (In-Cabin Benchmark)

Target Class	Instances	Precision	Recall	F1-Score	mAP@50	mAP@50-95
smoke (Smoking)	70	92.91%	93.58%	93.24%	96.89%	59.14%
using_phone	30	90.71%	86.67%	88.64%	90.41%	42.34%
earphone	122	90.43%	77.49%	83.47%	85.06%	35.82%
Overall Baseline (v1)	222	91.35%	85.91%	88.55%	90.78%	45.77%

2. Multi-Dataset Benchmark Synthesis

Dataset Benchmark	Evaluated Module	Sample Volume	Primary Accuracy / Recall	Status
YawDD Video Mirror	Drowsiness & MAR	10 videos (1,162 fr)	100.0% Video Accuracy / 100% Recall	PASSED
In-Cabin Benchmark	YOLO11s Distraction	201 images (222 box)	90.78% mAP@50 / 88.55% F1	PASSED
Kaggle State Farm	Out-of-domain Phone	1,065 images	99.70% Phone Catch Rate (331/332)	PASSED
Kaggle NTHU DDD	Slow-Blink & Nodding	250 images	86.00% Nodding / 99.2% YOLO clean	PASSED

3. Edge Hardware Latency & Deployment Recommendations

• **Primary Distraction Detector:** Keep Version 1 (weights/trained/yolo11s_distraction/best_openvino_model) as the production default. It provides **88.55% F1-score**, **96.89% smoking mAP@50**, and runs at **22.7 FPS** on CPU.
• **Secondary Balanced Model:** Retain Version 2 (weights/trained/yolo11s_balanced_distraction) for high-speed (**32.6 FPS**) and maximum phone precision (**95.05%**).
• **Temporal Filter:** CockpitSentinel's 2.0-second confirmation window filters out hand scratching and brief glances.
• **Team:** Satyapal Gaikwad, Darshan Patil, Yashodhan Jadhav — Mini Project 2026–27.